Exploratory Data Analysis of a Spotify Tracks Dataset

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Abstract

This report presents an Exploratory Data Analysis (EDA) of the Spotify Tracks dataset [1], which consists of 114,000 tracks across 114 genres with 21 features. EDA techniques used includes K-Means clustering, Principal Component Analysis (PCA), composite feature engineering, data visualisation, descriptive statistics, and data cleansing. PC1, which represents an acoustic/electric axis, accounts for 29.4% of the total variance and is heavily loaded by energy, loudness, and mood score. The silhouette analysis confirms that the main structure in the audio feature space is binary acoustic–electric by identifying k = 2 as the ideal number of clusters, producing one acoustic/low-energy cluster (n = 28,985) and one electric/high-energy cluster (n = 85,014).

1 Introduction

For each song in their catalogue, music streaming services like Spotify gather comprehensive acoustic metadata. These audio features, which are calculated measurements of attributes like danceability, energy, and valence, allow for data-driven analysis of musical content at scale. This report explores the following questions: How do audio features differ throughout the dataset and what correlations exist between them? Can dimensionality reduction reveal interpretable latent structure? Do unsupervised clusters correspond to meaningful musical groupings?

The dataset [1] was obtained from Kaggle and contains 114,000 tracks spanning 114 genres. It has 21 columns: 5 categorical or identifier fields (`track_id`, `artists`, `album_name`, `track_name`, `track_genre`) and 16 numerical features including audio properties and metadata.

2 Data Processing

2.1 Loading and Inspection

The dataset was loaded using `pd.read_csv("dataset.csv")` and inspected via `.columns`, `.dtypes`, `.iloc[100:110]`, `.describe()`, and `.info()`. The raw data contains 114,000 rows and 21 columns. Of these, 15 are numerical and 6 are categorical or identifier fields. The `explicit` column is boolean and `track_genre` is the categorical label for genre.

Table 1: Numerical audio features in the dataset.

Feature	Description
popularity	Spotify popularity score (0–100)
duration_ms	Track length in milliseconds
explicit	Whether track has explicit content
danceability	Suitability for dancing (0–1)
energy	Perceptual intensity (0–1)
key	Estimated musical key (0–11)
loudness	Overall loudness in dB
mode	Major (1) or minor (0)
speechiness	Presence of spoken words (0–1)
acousticness	Confidence of acoustic source (0–1)
instrumentalness	Absence of vocals (0–1)
liveness	Presence of live audience (0–1)
valence	Musical positiveness (0–1)
tempo	Estimated BPM
time_signature	Estimated beats per bar

2.2 Missing Values and Duplicates

Missing values were identified using `df.isnull().sum()`. Only three columns contained missing values: `artists`, `album_name`, and `track_name`, each with exactly 1 missing entry. These rows were removed using `df.dropna()`, leaving 113,999 rows. A duplicate check via `df.duplicated()` found 0 duplicate rows, so no further removal was necessary. The final clean dataset contains 113,999 tracks and 21 columns.

2.3 Descriptive Statistics

Table 2 summarises key descriptive statistics obtained from `df.describe()`. The mean popularity score is 33.2 (σ = 22.3), indicating a distribution skewed towards lower values, as most tracks are not chart hits. Mean danceability is 0.567 and mean energy is 0.642, both broadly consistent with a dataset dominated by contemporary popular music. The mean track duration is 3.80 min-

utes ($\sigma = 1.79$), with a maximum of 87.3 minutes indicating outliers (e.g. live recordings or podcasts).

Table 2: Descriptive statistics for key features ($n = 113{,}999$).

Feature	Mean	Std. Dev.	Median
popularity	33.24	22.31	35.00
danceability	0.567	0.174	0.580
energy	0.642	0.251	0.685
acousticness	0.314	0.332	0.172
valence	0.474	0.259	0.464
tempo	122.1	29.9	121.9
duration_min	3.801	1.788	3.548

3 Data Visualisation

3.1 Individual Feature Distributions

Plotly histograms were produced for popularity, danceability, energy, and valence. The popularity distribution is right-skewed, with the majority of tracks scoring below 50. The energy category breakdown confirms this: 54,445 tracks (47.8%) fall into the high energy category, 38,237 (33.5%) into medium energy, and 21,316 (18.7%) into low energy. Danceability and valence are approximately normally distributed, centred around 0.58 and 0.47 respectively.

A 3×3 Matplotlib subplot grid was produced for all nine core audio features. Speechiness and instrumentalness are heavily right-skewed, with the vast majority of tracks having values close to zero, showing that most tracks are lyrical and vocal. Liveness clusters tightly below 0.3, consistent with most tracks being studio recordings.

3.2 Correlation Heatmap

Over 11 numerical features were used to create a Pearson correlation heatmap. Loudness and energy have the biggest positive correlation: louder recordings are thought to be more energetic. Energy and loudness have a high negative correlation with acousticness, indicating that acoustic recordings are consistently quieter. There is a somewhat favourable association between danceability and vitality. Popularity exhibits weak connections with all audio characteristics, suggesting that no single sound quality significantly influences economic success.

3.3 Scatter Plots

Three bubble scatter plots were produced, each using a third feature to set point size:

- Popularity vs energy (size = $|\text{loudness}| \times 2$)
- Danceability vs valence (size = energy $\times$ 100)
- Acousticness vs energy (size = $|\text{loudness}| \times 2$)

The acousticness vs. energy scatter, which forms a wide diagonal band from high-acousticness/low-energy songs to low-acousticness/high-energy tracks, best illustrates the strong negative association between the two characteristics. In line with the heatmap's weak association, the popularity vs. energy scatter exhibits no discernible trend.

4 Feature Engineering

Four engineered features were added to the dataset:

Popularity label, The continuous popularity score was segmented into three bands using `pd.cut`: *low* (0–33), *medium* (34–66), and *high* (67–100). The resulting counts are: medium = 50,786 (44.5%), low = 39,562 (34.7%), and high = 7,632 (6.7%). The dominance of the medium band highlights the realistic long-tail distribution of streaming popularity.

Energy category, Energy was segmented into *low energy* (0–0.4), *medium energy* (0.4–0.7), and *high energy* (0.7–1.0). The dataset is majority high energy (47.8%), consistent with the mean energy of 0.642.

Mood score, A weighted composite capturing overall musical mood:

$$\text{mood_score} = 0.5 \cdot \text{valence} + 0.3 \cdot \text{danceability} + 0.2 \cdot \text{energy} \quad (1)$$

The mood score has mean 0.535, standard deviation 0.180, and ranges from 0 to 0.939, confirming coverage of the full emotional spectrum.

Duration in minutes, The raw `duration_ms` column was divided by 60,000. The mean track length is 3.80 minutes, with a median of 3.55 minutes and a maximum of 87.3 minutes. However, the maximum is an outlier in the dataset.

Plotly overlaid histograms and box plots were used to compare the engineered categorical groups. The box plots of popularity by energy category show that high-energy tracks do not enjoy a systematic popularity advantage, reinforcing the weak correlation seen in the heatmap.

5 PCA

PCA was applied to 12 standardised numerical features: `danceability`, `energy`, `loudness`, `speechiness`, `acousticness`, `instrumentalness`, `liveness`, `valence`, `tempo`, `duration_min`, `popularity`, and `mood_score`. All features were standardised using `StandardScaler` prior to PCA, as features operate on different scales (e.g. loudness in dB vs tempo in BPM).

5.1 Variance Explained

Table 3 shows the explained variance ratio per component. PC1 alone captures 29.4% of total variance, and PC2 a further 14.5%. 6 components are required to reach 80% cumulative variance, and 8 components to reach 90%, confirming moderate redundancy in the feature set arising from the correlated energy–loudness–acousticness group.

Table 3: PCA explained variance per component.

PC	Variance (%)	Cumulative (%)
1	29.4	29.4
2	14.5	43.9
3	10.4	54.3
4	9.0	63.3
5	8.0	71.3
6	7.3	78.6
7	7.1	85.7
8	6.5	92.1

5.2 PC1 Interpretation

The PC1 feature loadings (Table 4) show that `mood_score` (0.467), `loudness` (0.422), `energy` (0.410), and `valence` (0.371) have the largest positive loadings, while `acousticness` (-0.334) and `instrumentalness` (-0.257) have the largest negative loadings. PC1 can therefore be interpreted as an electric/upbeat vs. acoustic/instrumental axis: high PC1 = loud, energetic, and positive-sounding tracks; low PC1 = quiet, acoustic, and instrumental music.

Table 4: PC1 and PC2 feature loadings.

Feature	PC1	PC2
mood_score	0.467	0.301
loudness	0.422	−0.264
energy	0.410	−0.387
valence	0.371	0.414
danceability	0.299	0.387
tempo	0.134	−0.262
speechiness	0.085	0.006
liveness	0.052	−0.173
popularity	0.016	−0.030
duration_min	−0.043	−0.291
instrumentalness	−0.257	−0.117
acousticness	−0.334	0.414

PC1 vs PC2 scatter plots coloured by `energy_category` and `popularity_label` confirm that energy categories separate clearly along the PC1 axis, while popularity labels are uniformly distributed throughout the PCA space, consistent with the near-zero loading of popularity on PC1 (0.016).

6 Clustering

K-Means clustering was performed in the space of the top 4 principal components (63.3% of total variance). Using PCA-reduced features rather than raw features reduces noise and correlation effects.

6.1 Optimal Number of Clusters

Both the elbow method and silhouette scores were computed for $k \in \{2, \ldots, 10\}$. The silhouette scores are shown in Table 5. The score peaks at $k = 2$ (0.3184) and decreases uniformly thereafter, clearly identifying two clusters as the optimal solution.

Table 5: Silhouette scores for $k = 2$ to $k = 10$.

k	Silhouette Score
2	0.3184
3	0.2674
4	0.2566
5	0.2687
6	0.2484
7	0.2524
8	0.2431
9	0.2226
10	0.2267

6.2 Cluster Profiles

The final K-Means model ($k = 2$) was fit on the full dataset, producing Cluster 0 ($n = 28{,}985$, 25.4%) and Cluster 1 ($n = 85{,}014$, 74.6%). Mean feature values per cluster are shown in Table 6.

Table 6: Mean feature values per cluster.

Feature	Cluster 0	Cluster 1
Danceability	0.457	0.604
Energy	0.328	0.748
Valence	0.272	0.543
Acousticness	0.680	0.190
Instrumentalness	0.317	0.101
Tempo (BPM)	110.7	126.0
Popularity	32.3	33.6
Mood Score	0.339	0.602

Cluster 0 (Acoustic/Ambient): characterised by high acousticness (0.680), high instrumentalness (0.317), low energy (0.328), and a low mood score (0.339). This cluster captures acoustic, classical, and ambient tracks.

Cluster 1 (Electric/Rhythmic): characterised by high energy (0.748), low acousticness (0.190), high danceability (0.604), and a high mood score (0.602). This cluster captures pop, rock, electronic, and hip-hop.

Notably, the two clusters' mean popularity scores are nearly comparable (32.3 vs. 33.6), indicating that neither electric nor acoustic music is consistently more popular on Spotify. The danceability vs valence scatter plot coloured by cluster label visually confirms the separation, with Cluster 1 occupying the upper-right region (high danceability, high valence) and Cluster 0 the lower-left.

7 Conclusion

7.1 Summary of Findings

This EDA of the Spotify Tracks dataset demonstrated that audio features capture meaningful musical structure. The dominant pattern is acoustic/electric, confirmed by both the correlation heatmap (energy–loudness correlation) and PCA (PC1 loading pattern). Feature engineering added interpretable dimensions: the mood score (mean = 0.535) provides a compact scalar summary of a track's emotional character, and the energy category reveals that almost half of all tracks are classified as high energy.

Clustering with $k = 2$ produced an acoustic cluster (25.4% of tracks) and an electric cluster (74.6%), mirroring the PC1 axis. The near-equal popularity scores between clusters (32.3 vs 33.6) is a key finding: despite their very different audio characteristics, acoustic and electric tracks attract largely similar levels of streaming engagement, suggesting that listener preference is diverse and popularity is not driven by just energy or mood.

7.2 Limitations

The dataset spans 114 genres with unequal representation; some genres may have too few tracks to draw reliable information. The maximum duration of 87.3 minutes is a clear outlier, likely representing a live recording or podcast mislabelled as a music track. Spotify doesn't publicly explain how audio features such as danceability and energy are calculated, limiting the ability to interpret properly.

7.3 Reflection on Techniques

The PCA was particularly valuable for understanding feature redundancy: 6 components being required for 80% variance determines the moderate (not extreme) correlation structure of the dataset. The silhouette analysis provided an objective benchmark for selecting $k = 2$, avoiding the subjectivity of just the elbow plot.

References

[1] M. Pandya. Spotify Tracks Dataset, 2022. https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset